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Are pre-opening periods important? Evidence from Chinese market lunch breaks

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ABSTRACT

The Chinese stock market design offers a unique opportunity to examine the impact of pre-open call auctions on price efficiency and price discovery. By comparison, we find that opening prices of morning sessions, as determined by call auctions, are more efficient and informative than prices of the afternoon session, as determined by a continuous auction mechanism. Using corporate announcements to compare the speed of price adjustments in response to overnight information versus lunch break information, we find that prices react much more rapidly to overnight announcements, consistent with pre-open call auctions facilitating price discovery to announcements. Results also suggest that corporate announcements enhance stock liquidity with the effect of overnight announcements on liquidity being stronger than the effects of lunch break announcements. Overall, our evidence highlights the utility of pre-open call auctions.

1. Introduction

A fundamental issue in market microstructure is the formation of prices in financial markets. The introduction of a continuous auction mechanism offers a trading environment to generate a fair price for an asset and translate investors' latent demand into realized volumes (Theissen, 2000; Han et al., 2022). However, establishing an efficient opening price after market closure, which may be due to the close of regular trading, trading suspensions, or temporary halts, is challenging. Openings are an especially crucial period because uncertainty regarding fundamental values is high following market closures (Madhavan and Panchapagesan, 2000), and may influence trader disposition into the subsequent trading day.

As a trading mechanism, call auctions determine a fair execution price by batching buy and sell orders into a single price, with the resulting maximizing of the number of shares traded (Schwartz and Davis, 2010). Markets around the world have incorporated or are planning to incorporate, a pre-open call auction mechanism into their design. Whether financial markets need such a mechanism has become an increasingly important topic. Thus, the object of this paper is to examine the impact of such pre-open call auctions on price efficiency and the price discovery process.

The introduction of pre-open call auctions has attracted considerable academic interest (e.g., Madhavan, 1992; Domowitz and Madhavan, 2001). However, scholars have not reached unanimous consent on whether the introduction of pre-open call auctions can enhance market quality. A number of studies provide strong support for the utility of pre-open call auctions (Pagano and Schwartz, 2003; Comerton-Forde et al., 2007a; Gerace et al., 2015; Agarwalla et al., 2015). This literature argues that a well-designed call auction

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pre-open period facilitates learning about the fundamental values of assets prior to trading. Thus, the pre-open period can reduce price volatility, transaction cost, and asymmetric information and contribute to the formation of efficient opening prices (Comerton-Forde et al., 2007a; Chang et al., 2008; Comerton-Forde and Rydger, 2006; Barclay and Hendershott, 2008; Cole et al., 2022; Shi et al., 2023). Moreover, pre-open call auction periods account for a significant proportion of daily trade (Madhavan and Panchapagesan, 2000; Kehr et al., 2001), the pre-open call auctions also play an important role in incorporating information into the opening price through investor learning (Biais et al., 1999; Cao et al., 2000; Moshirian et al., 2012).

Another piece of literature contradicts the utility of pre-open call auctions for facilitating open trading. Ellul et al. (2005) find that pre-open call auctions may not reduce trading costs. Ibikunle (2015) finds that pre-open call auctions only provide highly efficient opening prices for the largest volume stocks but not for other stocks. Comerton-Forde et al. (2007b) find that the design of the Hong Kong Exchange and Clearing Ltd. differs significantly from opening auctions in other markets, indicating a decrease in the market quality following the introduction of the opening call auction. It remains unclear whether call auctions are beneficial to financial markets. In this paper, we attempt to answer two primary questions about the pre-open call auction mechanism on price efficiency. First, whether the pre-open call auction mechanism help the market to establish opening prices with less noise and more efficiency? Second, whether the pre-open call auction mechanism affect market reaction to corporate announcements?

As the largest equity market in emerging economies, the Chinese stock market has attracted investment from foreign institutional investors by offering the possibility of asset diversification and exerting a significant influence on the global capital market (Peng et al., 2024). Additionally, several mechanism designs in the Chinese stock market, such as daily price limits (Zhang et al., 2022), IPO price limits (Wang et al., 2023), Shanghai-Hong Kong Stock Connect program (Chen et al., 2023), circuit breakers (Wang et al., 2019; Li et al., 2021), etc., provide the opportunities to carry out quasi-natural experiments for academic research. More importantly, Chinese stock exchanges also incorporate call auctions to determine daily opening and closing prices. However, each trading day has two continuous auction sessions: a morning session and an afternoon session. These sessions are split by a lunch break. There are no call auctions used to close morning sessions or to open the afternoon sessions. This unique mechanism design in the Chinese stock market allows us to examine whether pre-open call auctions are important for the financial market by comparing market functioning between morning and afternoon sessions.

The first part of this study focuses on the impact of pre-open call auctions on opening price efficiency. This issue stems from the mixed evidence in prior literature regarding whether financial markets need call auctions. We attempt to answer this question by comparing the difference in opening price efficiencies between morning and afternoon sessions.

We initially employ the weighted price contribution analysis to examine the price discovery process for opening prices in the morning session and the afternoon session. We find that the pre-open call auction mechanism accelerates more information incorporated in the opening price. Subsequent the extent to which opening prices reflect noise or information is another key issue. The unbiasedness regression analysis allows us to test the signal-to-noise ratio of the opening price. We find that the opening price of the morning session is less noisy than that of the afternoon session. By comparison, we provide empirical evidence to support that pre-open call auctions can improve opening price efficiency.

The second part of this study focuses on the impact of pre-open call auctions on the market reaction to corporate announcements. As one kind of the most important public information events, how the market reacts to corporate announcements is a fundamental issue in the financial field. To ensure all relative information is fully accurate, with timely disclosure, firms are also allowed to release news announcements during lunch break periods. Over 2.2 % of corporate announcements in our sample are released during lunch break periods.

The unique time distribution of corporate announcements allows us to compare the differences in price discovery between reactions to corporate announcements released during overnight and lunch break periods. Portfolio analysis shows that stock prices adjust rapidly to overnight announcements, with most price adjustments in response to overnight announcements occurring within five minutes after openings of morning sessions. This result shows that the pre-open call auction would accelerate the price adjustment to the corporate announcement.

Additionally, liquidity is also crucial in the financial market because it enhances efficiency and reduces the risk of price manipulation. We further examine the impact of overnight announcements and lunch break announcements on stock liquidity, finding that corporate announcements enhance stock liquidity, with this enhancing effect of announcements on stock liquidity being much stronger for overnight announcements than for lunch break announcements.

We contribute to the literature as the nature of our data and the institutional settings of the Chinese stock market provide a unique perspective for exploring a number of issues regarding the utility of pre-open call auctions. Existing literature has not yet reached a consensus on the effect of the pre-open call auction mechanism on market quality (Comerton-Forde et al., 2007a; Agarwalla et al., 2015; Chang et al., 2008; Comerton-Forde et al., 2007b; Ibikunle, 2015; Ellul et al., 2005). The unique design of the Chinese stock market allows us to examine the impact of pre-open call auctions on price efficiency and price discovery by comparing the differences between the morning and afternoon sessions. Our results show that the opening prices of the morning sessions are more efficient and informative than those of the afternoon sessions. Our paper complements the literature by supporting pre-open call auctions as important for opening price efficiency.

We also provide insights into the differences in market reactions in response to overnight announcements versus lunch break announcements. A large body of literature focuses on differences in price discovery in response to daytime versus overnight news announcements (Greene and Watts, 1996; Gerlach and Lee, 2011; Moshirian et al., 2012; Jain et al., 2019). However, little research

explores the price discovery process with respect to lunch break announcements. We provide empirical evidence showing that price adjustments to lunch break announcements are slower than for overnight announcements, with enhancing effects on liquidity also relatively weaker. We provide new evidence for the role of pre-open call auctions in facilitating the price discovery process, highlighting the prospect of further enhancing this role by timing the release of corporate announcements to occur during overnight periods. More broadly, our paper adds to the literature on understanding the price discovery process after public announcements (Dai et al., 2022; Wong et al., 2021; Wang et al., 2023).

The remainder of this study is organized as follows. Section 2 describes the institutional design of the Chinese stock market. Section 3 describes the data collection process and variable construction. Section 4 reports the main results and discussion. Section 5 concludes.

2. Institutional details

The Chinese stock market is a fully automated order-driven electronic trading system, constructed via two mainland exchanges: the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE). According to the trading methodology of the Chinese stock market, the SSE and SZSE operate a continuous market five days a week, from 9:30 am to 14:57 pm, with a lunch break from 11:30 am to 13:00 pm. Before the morning trading session, these two exchanges operate pre-open call auctions from 9:15 am to 9:30 am to determine opening prices. 9:15 am and 9:25 am orders may be entered, amended, and withdrawn, but not executed. Between 9:25 am to 9:30 am, there is a non-cancellation period during which orders can be entered, but not amended or withdrawn. At 9:30 am, overlapping orders are matched at a single price. However, there is no such procedure before the afternoon session. After the normal trading session, the pre-close call auctions operate between 14:57 pm and 15:00 pm to determine the closing price.

Auction trading of securities in the Chinese market is conducted either as continuous auctions¹ or call auctions.² For normal trading sessions, the continuous auction method is used to match orders. All orders are continuously one-by-one matched and executed based on the principles of price priority and time priority. For the opening and closing price determination, a call auction method is used to determine the opening and closing prices. All orders are matched at a single price, which should be determined according to three principles: (1) a price that will produce the highest trading volume; (2) a price that allows all buy orders with the highest bid and all sell orders with a lower ask to be executed; (3) a price that allows all buy orders at that price to be executed or all sell orders at that price to be executed. If more than one price satisfies the conditions above, the algorithm applies another principle as a supplement, which selects a price that minimizes the unmatched volume.

Chinese traders participate in a daily compulsory lunch break of one and a half hours from 11:30 am to 13:00 pm. Such imposed lunch breaks are rather uncommon. Among global stock exchanges, only a few Asia-Pacific exchanges have trading lunch breaks, although these comprise the most important Asian exchanges. Tokyo, Hong Kong, Shanghai, Shenzhen, and Singapore, take one- to two-hour breaks each trading day. However, unlike the Tokyo stock exchange where the opening price of the afternoon session is generated from a call auction during the pre-open period, Chinese stock exchanges have no pre-open call auctions to determine the opening price of afternoon sessions. Rather, afternoon sessions are opened with a continuous auction.

To improve market information transparency and lower the information acquisition cost, listed firms should prepare and disclose periodic reports and adhere to administrative regulations, rules of competent authorities, and other regulatory documents. In the case of material matters that may considerably affect the trading price of a listed firm's share and that are not yet known to the investors, listed firms shall file with the Exchange and disclose their ad hoc reports in a timely manner. In particular, listed firms are required to submit their reports to the stock exchange (SSE or SZSE) and release them on the medium designated by the China Securities Regulatory Commission (CSRC). In the Chinese stock market, most corporate public news items are released during the overnight period (15:00–23:59 and 00:00–09:30). But to ensure the timeliness of information disclosure, some listed firms release public news during the normal trading period (09:30–11:30 and 13:00–15:00) and lunch break period (11:30–13:00).

As shown in Fig. A1 in Appendix A, there were 1,637,764 corporate announcements published during our period 1,598,485 announcements (97.60 %) from 3562 listed firms are released during the overnight period of 553,347 firm-days. 36,051 announcements (2.201 %) from 2033 listed firms are released during the lunch break period of 22,068 firm-days. 3255 announcements (0.199 %) are released from 976 listed firms during the normal trading period of 2865 firm-days. Table A1 reports the distribution by type of corporate announcements released during these three periods. The type distribution of corporate announcements released during the lunch break period is similar to that during the overnight period.

3. Data and variables

We construct our sample based on a variety of data sources. First, we obtain corporate announcement data from the Chinese Stock Market and Accounting Research Database (CSMAR), covering all firms listed in the Chinese stock market from January 1, 2013 to October 31, 2017.³ We collect limit order book data and tick-by-tick transaction data from Chinese stock exchanges and obtain 1-min high-frequency price data from the RESSET database.

To better understand the difference in price discovery of overnight announcements and lunch break announcements, our sample

¹ Continuous auction refers to the process of continuous matching of buy and sell orders on a one-by-one basis.

² Call auction refers to the process of one-time centralized matching of buy and sell orders accepted during a specified period of time.

³ We don't adopt an extensive period because of the size of order book data and tick-by-tick transaction data are prohibitively large.

consists of listed firms with at least one announcement released during the lunch break period of effective firm-days,⁴ and excludes the firm-days with announcements released in the normal trading period. Our final sample consists of 1,749,520 effective firm-days from 1934 listed firms, including 274,328 effective firm-days with at least one announcement released during the overnight period and 17,769 effective firm-days with at least one announcement released during the lunch break period.

3.1. Liquidity

To compare the impact of overnight announcements and lunch break announcements on market quality, we focus on market liquidity. Many methods have been used in prior studies to measure market liquidity, including the price-based method, volume-based method, price-volume-based methods, and time-based methods (O'Hara, 1995; Liu et al., 2021). Given that liquidity is a multiple-dimensional concept and cannot be assessed by one indicator alone, three variables are used.

The first liquidity variable is quote spread (QS). Following Bessembinder and Kaufman (1997), quote spread is calculated as the difference between the best bid price and the best ask price, scaled by the quote midpoint, which is the average of the best bid price and the best ask price. More specifically, quote spread is defined as follows:

$$QS_t = \frac{Ask_t - Bid_t}{(Ask_t + Bid_t)/2} \quad (1)$$

where Ask_t denote the best ask price and Bid_t denote the best bid price at time t .

The opportunity to transact inside the bid-ask spread is another measure of liquidity as it evaluates how close to the “best price”, which is proxy by the midpoint of the bid and ask prices., is possible for trade. Effective spread (ES) usually is considered to be a more realistic indicator of market liquidity than quote spread. We calculate the effective spread as the absolute value of the difference between the actual trade price and the quote midpoint scaled by the quote midpoint. More specifically, the effective spread is defined as follows:

$$ES_t = \frac{|P_t - (Ask_t + Bid_t)/2|}{(Ask_t + Bid_t)/2} \quad (2)$$

where P_t denotes the actual price.

Quote spread and effective spread are measured through the best bid and best ask prices, which overlook the order depth in the limit order book. We consider the third market liquidity measure: relative market depth (*Depth*). Relative market depth is a liquidity indicator based on the quantity dimension, which complements the price dimension (spread) measure of liquidity and reflects liquidity from the perspective of liquidity supply, and represents the ability of the optimal quantity of order available to absorb the trading volume. Following Liu et al. (2021), we calculate the relative market depth as follows:

$$Depth_t = \frac{100}{N \times TV_t} \sum_{m=1}^N (V_{t,m}^{ask} + V_{t,m}^{bid})/2 \quad (3)$$

where m is a shorter period within time t , N is the number of periods in each time t , $V_{t,m}^{ask}$ is the quantity of orders at the best ask price, and $V_{t,m}^{bid}$ is the quantity of orders at the best bid price. TV_t is the trading volume, which is multiplied by 100 to ensure the numerator and the denominator match in magnitude, we aggregate the three liquidity measures for each firm and each 15-min interval.

3.2. Other variables

Based on market liquidity literature (Chordia et al., 2001; Chung and Chuwonganant, 2014), we select a number of control variables for inclusion in our analyses. We include return volatility (*Volt*), calculated as the standard deviation of stock return based on the limit order book in a given period. Return volatility serves as a proxy for the level of uncertainty about stock value.

Order size (*OS*), trading volume (*Volume*), and trade frequency (*TF*) are three indicators of order execution quality. Consequently, we control for order size, which is calculated as the average number of shares traders submitted in a given period. We also control trading volume, which is calculated as the total number of shares executed in a given period, and trade frequency, which is calculated as the average number of trades in a given period.

4. Results and discussions

4.1. Overview of market activities during the morning session and afternoon session

We provide an overview of market activities during the morning session and the afternoon session in Table 1–2. Table 1 presents a summary of descriptive statistics on market liquidity measures. Table 2 presents summary descriptive statistics on the volatility, order

⁴ Effective firm-days refer to the normal trading firm-days because of some firms are suspended in trading days.

Table 1
Summary statistics of market liquidity.

Period		P1	P2	P3	P4	P5	P6	P7	P8
Panel A: Entire Sample									
ES	Morning Session (M)	0.001014	0.0007330	0.0007007	0.0006882	0.0006876	0.0006942	0.0007018	0.0007097
	Afternoon Session (A)	0.0007170	0.0006832	0.0006741	0.0006709	0.0006692	0.0006661	0.0006547	0.0006377
	(A-M)/M	−0.2044***	−0.0265***	0.0068***	0.00231***	0.0255***	0.0146***	−0.0121***	−0.0475***
	z-stats	(−871.14)	(−98.15)	(24.45)	(80.89)	(89.93)	(51.19)	(−44.03)	(−168.16)
QS	Morning Session (M)	0.001813	0.001421	0.001367	0.001349	0.001352	0.001367	0.001384	0.001402
	Afternoon Session (A)	0.001413	0.001348	0.001329	0.001323	0.001318	0.001310	0.001283	0.001203
	(A-M)/M	0.1717***	−0.0103***	0.0152***	0.0261***	0.0240***	0.0088***	−0.0231***	−0.1000***
	z-stats	(−746.27)	(−39.26)	(56.49)	(95.27)	(87.77)	(32.64)	(−89.45)	(−402.62)
Depth	Morning Session (M)	0.03010	0.04101	0.05168	0.06299	0.07313	0.08166	0.08881	0.09929
	Afternoon Session (A)	0.09879	0.09608	0.09064	0.09011	0.08499	0.08030	0.06905	0.06257
	(A-M)/M	2.8341***	1.6848***	1.1076***	0.7924***	0.5298***	0.3389***	0.1103***	0.0429***
	z-stats	(1125.21)	(971.77)	(814.06)	(678.01)	(529.34)	(382.80)	(152.97)	(61.14)
Panel B: Overnight Announcement Sample									
ES	Morning Session (M)	0.0009767	0.0007159	0.0006855	0.0006724	0.0006724	0.0006870	0.0006936	0.0007027
	Afternoon Session (A)	0.0006694	0.0006607	0.0006571	0.0006571	0.0006556	0.0006524	0.0006413	0.0006236
	(A-M)/M	−0.2016***	−0.0247***	0.0088***	0.0254***	0.0282***	0.0174***	−0.0096***	−0.0454***
	z-stats	(−325.80)	(−34.51)	(11.91)	(33.58)	(36.96)	(22.86)	(−12.93)	(−59.69)
QS	Morning Session (M)	0.001744	0.001375	0.001327	0.001309	0.001313	0.001328	0.001346	0.001361
	Afternoon Session (A)	0.001377	0.001313	0.001295	0.001288	0.001283	0.001275	0.001249	0.001164
	(A-M)/M	−0.1657***	−0.0063***	0.0185***	0.0291***	0.0271***	0.0116***	−0.0208***	−0.1025***
	z-stats	(−272.04)	(−9.04)	(25.83)	(40.04)	(37.27)	(16.07)	(−29.98)	(−155.54)
Depth	Morning Session (M)	0.02460	0.03604	0.04632	0.05710	0.06669	0.07492	0.08174	0.09069
	Afternoon Session (A)	0.08896	0.08791	0.08350	0.08325	0.07856	0.07422	0.06405	0.05834
	(A-M)/M	3.2364***	1.8005***	1.1537***	0.8122***	0.5417***	0.3429***	0.1063***	−0.0627***
	z-stats	(444.06)	(383.95)	(320.25)	(266.59)	(206.61)	(148.44)	(56.79)	(33.19)
Panel C: Lunch Break Announcement Sample									
ES	Morning Session (M)	0.001044	0.0007214	0.0006860	0.0006673	0.0006691	0.0006766	0.0006856	0.0006944
	Afternoon Session (A)	0.0007413	0.0006594	0.0006490	0.0006440	0.0006436	0.0006375	0.0006251	0.0006017
	(A-M)/M	−0.1750***	−0.0327***	0.0020	0.0243***	0.0229***	0.0099***	−0.0192***	−0.0632***
	z-stats	(−55.48)	(−9.78)	(0.58)	(6.75)	(6.27)	(2.76)	(−5.57)	(−17.72)
QS	Morning Session (M)	0.001846	0.001391	0.001333	0.001305	0.001311	0.001330	0.001349	0.001368
	Afternoon Session (A)	0.001442	0.001293	0.001273	0.001264	0.001263	0.001249	0.001219	0.001124
	(A-M)/M	−0.1475***	−0.0188***	0.0088***	0.0241***	0.0206***	0.0017	−0.0335***	−0.1205***
	z-stats	(−47.93)	(−5.75)	(2.59)	(6.98)	(5.86)	(0.50)	(−10.33)	(−38.60)
Depth	Morning Session (M)	0.02447	0.03231	0.04070	0.05012	0.05660	0.06359	0.06836	0.07782
	Afternoon Session (A)	0.06121	0.06419	0.06307	0.06354	0.06074	0.05834	0.05134	0.04981
	(A-M)/M	1.9312***	1.3591***	0.9174***	0.6338***	0.4411***	0.2757***	0.0762***	0.0347***
	z-stats	(78.58)	(75.09)	(64.19)	(52.36)	(40.60)	(28.58)	(9.39)	(4.25)
Panel D: Non-Announcement Sample									
ES	Morning Session (M)	0.001021	0.0007361	0.0007036	0.0006912	0.0006905	0.0006972	0.0007046	0.0007127
	Afternoon Session (A)	0.0007193	0.0006859	0.0006768	0.0006736	0.0006719	0.0006687	0.0006573	0.0006406
	(A-M)/M	−0.2052***	−0.0267***	0.0065***	0.00226***	0.0250***	0.0141***	−0.0125***	−0.0477***
	z-stats	(−806.36)	(−91.31)	(21.54)	(73.29)	(80.67)	(45.74)	(−42.01)	(−156.14)
QS	Morning Session (M)	0.001825	0.001429	0.001374	0.001356	0.001359	0.001374	0.001391	0.001409
	Afternoon Session (A)	0.001419	0.001355	0.001336	0.001329	0.001325	0.001317	0.001289	0.001211
	(A-M)/M	−0.1730***	−0.0109***	0.0147***	0.0256***	0.0235***	0.0084***	−0.0235***	−0.0994***
	z-stats	(−693.69)	(−38.30)	(50.37)	(86.20)	(79.33)	(28.64)	(−83.79)	(−368.93)
Depth	Morning Session (M)	0.03111	0.04197	0.05273	0.06416	0.07444	0.08303	0.09027	0.1010
	Afternoon Session (A)	0.1009	0.09785	0.09219	0.09159	0.08636	0.08161	0.07012	0.06344
	(A-M)/M	2.7743***	1.6683***	1.1016***	0.7906***	0.5285***	0.3389***	0.1114***	0.0395***
	z-stats	(1031.62)	(888.73)	(744.97)	(620.66)	(485.17)	(351.37)	(141.72)	(52.06)

This table reports the descriptive statistics regarding the market liquidity in eight 15-min intervals during the morning session (09:30 am to 11:30 am) and afternoon session (13:00 pm. to 15:00 pm.), respectively. QS denotes the quote spread, which is calculated as the difference between the best bid price and the best ask price, divided by the quote midpoint. ES denotes the effective spread, which is calculated as the difference quote midpoint and the actual transaction price divided by the quote midpoint. Depth denotes the quoted depth, which indicates market liquidity from the perspective of liquidity supply and represents the ability of the optimal number of orders available for exchange in the market. The differences in liquidity between the morning session and afternoon session are presented below. A t-statistic is calculated to test for a statistically significant difference in the morning session and the afternoon session liquidity estimates. Panel A reports the results for the entire sample. Panel B reports the results for the overnight announcement sample, which consists of firm-days with at least one announcement released on the overnight period and without any announcement released on any other period. Panel C reports the results for the lunch break announcement sample, which consists of firm-days with at least one announcement released on the lunch break period and without any announcement released on any other period. Panel D reports the results for the non-announcement sample, which consists of firm-day without any announcement released on that day.

Table 2

Summary descriptive statistics of the market environment.

Period		P1	P2	P3	P4	P5	P6	P7	P8
Panel A: Return Volatility (Volt)									
Entire Sample	Morning Session (M)	0.001122	0.0007702	0.0007230	0.0007020	0.0006968	0.0006997	0.0007035	0.0007123
	Afternoon Session (A)	0.0007345	0.0006818	0.0006686	0.0006645	0.0006601	0.0006562	0.0006437	0.0006121
	(A-M)/M	−0.2646***	−0.0410***	0.0020**	0.0266***	0.0410***	0.0450***	0.0205***	−0.0506***
	z-stats	(−802.13)	(−94.44)	(2.09)	(61.47)	(6.63)	(3.67)	(3.27)	(−52.19)
Overnight Announcement Sample	Morning Session (M)	0.001113	0.0007478	0.0007028	0.0006809	0.0006759	0.0006783	0.0006837	0.0006901
	Afternoon Session (A)	0.00071	0.0006600	0.0006474	0.0006432	0.0006386	0.0006331	0.0006204	0.0005853
	(A-M)/M	−0.2770***	−0.0441***	−0.0036***	0.0229***	0.0735*	0.0211***	0.0357	−0.0654***
	z-stats	(−367.28)	(−30.26)	(−3.37)	(20.90)	(1.79)	(19.07)	(1.08)	(−50.87)
Lunch Announcement Sample	Morning Session (M)	0.001194	0.0007856	0.0007317	0.0006988	0.0006959	0.0007027	0.0007029	0.0007146
	Afternoon Session (A)	0.0008436	0.0006765	0.0006577	0.0006497	0.0006438	0.0006373	0.0006234	0.0005850
	(A-M)/M	−0.1798***	−0.0471***	−0.0138***	0.0175***	0.0201***	0.0150***	0.2036	−0.0771***
	z-stats	(−39.51)	(−8.99)	(−2.89)	(3.50)	(3.86)	(2.78)	(0.96)	(−13.88)
No Announcement Sample	Morning Session (M)	0.001123	0.0007740	0.0007266	0.0007058	0.0007006	0.0007035	0.0007071	0.0007162
	Afternoon Session (A)	0.0007373	0.0006858	0.0006725	0.0006685	0.0006642	0.0006606	0.0006481	0.0006172
	(A-M)/M	−0.2633***	−0.0403***	0.0032***	0.0273***	0.0357***	0.0494***	0.0163***	−0.0477***
	z-stats	(−720.91)	(−90.69)	(2.81)	(57.82)	(18.27)	(3.40)	(3.76)	(−42.42)
Panel B: Order Size (OS)									
Entire Sample	Morning Session (M)	3476.6485	7364.4899	10,307.2470	12,663.8896	14,637.1769	16,370.1018	17,883.8216	19,181.3384
	Afternoon Session (A)	20,990.4722	22,656.4208	24,404.1210	26,091.9053	27,904.1152	29,850.3471	32,255.4281	35,447.1435
	(A-M)/M	4.9085***	2.0937***	1.4265***	1.2582***	1.5667***	1.4393***	2.2981***	4.3400***
	z-stats	(1593.60)	(847.71)	(93.51)	(24.56)	(4.37)	(12.46)	(4.78)	(2.09)
Overnight Announcement Sample	Morning Session (M)	3813.5451	8191.7031	11,441.3990	14,033.7076	16,195.4711	18,093.0133	19,754.5095	21,175.5314
	Afternoon Session (A)	23,183.9631	24,975.8728	26,880.0924	28,717.8016	30,699.8621	32,826.7704	35,455.1297	38,978.2829
	(A-M)/M	4.9578***	2.0626***	1.4718***	1.5745***	1.4525***	2.2731***	7.3747**	19.6431***
	z-stats	(1114.42)	(501.90)	(21.00)	(7.09)	(5.63)	(5.11)	(2.32)	(14.86)
Lunch Announcement Sample	Morning Session (M)	3597.5402	7633.6849	10,653.5358	13,066.0285	15,080.6838	16,863.7002	18,410.5137	19,740.6532
	Afternoon Session (A)	22,066.2434	23,826.6769	25,526.1014	27,173.9087	29,007.6213	30,986.8517	33,445.9886	36,746.0693
	(A-M)/M	5.0130***	2.1361***	1.4179***	1.1141***	2.1204*	0.8602***	0.8663***	0.9168***
	z-stats	(602.52)	(571.70)	(509.97)	(153.17)	(1.84)	(439.24)	(30.49)	(20.71)
No Announcement Sample	Morning Session (M)	3415.5160	7214.5117	10,101.9101	12,416.0459	14,355.3536	16,058.5516	17,545.6399	18,820.8452
	Afternoon Session (A)	20,588.9636	22,231.4994	23,952.0335	25,613.5608	27,395.3167	29,309.1606	31,673.9640	34,805.5963
	(A-M)/M	4.8985***	2.0988***	1.4062***	1.2044***	1.5375***	1.2964***	1.4207***	1.6894***
	z-stats	(1371.61)	(738.56)	(334.58)	(25.83)	(3.65)	(11.51)	(11.69)	(8.70)
Panel C: Volume									
Entire Sample	Morning Session (M)	15,637.5614	11,604.8573	9480.4213	8038.5791	7101.3191	6356.9834	5951.2561	5463.8108
	Afternoon Session (A)	5762.2859	5870.2385	6079.0773	6086.3683	6357.1201	6774.8727	8031.6202	10,074.8477
	(A-M)/M	−0.3418***	−0.2000***	−0.0118	0.2731***	0.8245***	0.8838***	1.4147***	2.2834***
	z-stats	(−37.71)	(−26.18)	(−1.41)	(13.29)	(5.28)	(28.89)	(22.90)	(41.53)
Overnight Announcement Sample	Morning Session (M)	23,165.8761	15,215.4537	11,947.8924	9927.5864	8666.8680	7687.4983	7147.4691	6569.6202
	Afternoon Session (A)	7004.6198	7005.2705	7239.7356	7191.3390	7533.8528	8023.4721	9503.3580	12,045.0632
	(A-M)/M	−0.4372***	−0.2208***	−0.0445*	0.3151***	1.4372***	1.2765***	2.2045***	2.6159***
	z-stats	(−54.53)	(−10.04)	(−1.83)	(6.69)	(2.66)	(8.67)	(5.92)	(22.56)
Lunch Announcement Sample	Morning Session (M)	12,995.3222	9575.7021	7753.3209	6477.2206	5688.2366	5190.6910	4846.8308	4414.1708
	Afternoon Session (A)	8722.0202	5888.3739	5501.6012	5309.1140	5392.7813	5625.5932	6803.2663	8185.1440
	(A-M)/M	−0.3519***	−0.0431*	−0.2065***	0.9610*	5.1560	1.3945***	1.4216***	2.3676***

(continued on next page)

Table 2 (continued)

Period		P1	P2	P3	P4	P5	P6	P7	P8
No Announcement Sample	z-stats	(−4.21)	(−1.68)	(−3.41)	(1.79)	(1.13)	(3.91)	(10.95)	(21.78)
	Morning Session (M)	14,368.0259	11,001.6228	9072.7962	7728.3894	6844.5327	6138.5393	5755.5654	5283.4424
	Afternoon Session (A)	5515.1074	5672.9688	5884.1140	5902.4250	6164.2589	6571.6141	7791.0941	9754.4615
	(A-M)/M	−0.3320***	−0.1987***	−0.0041***	0.2596***	0.6781***	0.8104***	1.2756***	2.2239***
	z-stats	(−31.20)	(−24.26)	(−0.46)	(11.90)	(4.42)	(32.06)	(38.55)	(35.87)
Panel D: Trading Frequency (TF)									
Entire Sample	Morning Session (M)	980.7653	732.5597	596.8867	506.5081	446.9790	400.6095	373.2184	341.3645
	Afternoon Session (A)	371.0714	374.3521	385.3265	383.5563	400.1612	424.2766	496.4348	615.0974
	(A-M)/M	−0.4280***	−0.2994***	−0.1273***	0.0422***	0.2424***	0.4546***	0.8236***	1.4985***
	z-stats	(−456.08)	(−158.84)	(−58.66)	(10.39)	(34.87)	(11.14)	(233.94)	(387.56)
Overnight Announcement Sample	Morning Session (M)	1403.6914	940.9790	739.4415	615.9607	539.2169	478.6107	443.1571	405.4734
	Afternoon Session (A)	444.4825	440.7691	452.6689	448.3729	468.2567	496.7669	580.3682	727.3335
	(A-M)/M	−0.4983***	−0.3230***	−0.1406***	0.0377***	0.2687***	0.4886***	0.8713***	1.5490***
	z-stats	(−214.33)	(−30.31)	(−18.71)	(3.77)	(9.35)	(29.61)	(51.22)	(100.68)
Lunch Announcement Sample	Morning Session (M)	935.1626	697.3010	563.9822	472.1130	413.8246	376.8482	350.4053	318.3369
	Afternoon Session (A)	625.0093	439.9200	407.1229	389.7639	392.8430	408.9117	485.4016	585.8127
	(A-M)/M	−0.0857***	−0.1150***	0.0136	0.2176***	0.4427***	0.5912***	0.9297***	1.6744***
	z-stats	(−5.01)	(−8.73)	(0.82)	(5.58)	(4.18)	(12.67)	(29.72)	(38.74)
No Announcement Sample	Morning Session (M)	907.9131	696.6028	572.3259	487.7518	431.2543	387.2130	361.2244	330.4093
	Afternoon Session (A)	355.5548	362.0461	373.3363	372.1419	388.3099	411.7966	481.9045	595.9528
	(A-M)/M	−0.4208***	−0.2970***	−0.1264***	0.0412***	0.2358***	0.4472***	0.8137***	1.4874***
	z-stats	(−412.07)	(−244.24)	(−57.32)	(9.24)	(36.53)	(113.91)	(281.55)	(404.30)

This table reports the descriptive statistics of the market environment in eight 15-min intervals during the morning session (09:30 am to 11:30 am) and the afternoon session (13:00 pm to 15:00 pm), respectively. The differences in liquidity between the morning session and the afternoon session are presented below. A z statistic is calculated to test for a statistically significant difference in the morning session and the afternoon session environment measures. The entire sample consists of all firm-days in our sample; the overnight announcement sample consists of firm-days with at least one announcement released on the overnight period and without announcement released on any other period; the lunch break announcement sample consists of firm-days with at least one announcement releasing on lunch break period (11:30 am to 13:00 pm), and without announcement released on any other period.; non-announcement sample consists of firm-days without any announcement released on that day. Panel A reports the descriptive statistics for return volatility, which is calculated as the standard deviation of the return of each transaction in a given period. Panel B reports the descriptive statistics for order size, which is calculated as the average order size in a given period. Panel C reports the descriptive statistics for volume, which is calculated as the total transaction volume in a given period. Panel D reports the descriptive statistics for transaction frequency, which is calculated as the total number of transactions in a given period.

size, trading volume, and trade frequency. To consider different market information environments, we provide results for the entire sample, the subsample of firm-day with at least one overnight announcement and without any lunch break announcement (overnight announcement sample), the subsample of firm-day with at least one lunch break announcement, and without any overnight announcement (lunch break announcement sample), and the subsample of firm-day without any announcement (non-announcement sample). When comparing market activities between the morning session and the afternoon session, we use the relative difference to eliminate the influence of firm-level factors. Specifically, we calculate the relative difference as the ratio of market activities during the afternoon session and the morning session, and minus 1.⁵

In Panel A of Table 1, we report the liquidity across 15-min intervals during the morning session and the afternoon session for the entire sample. We find that there is an intraday L-shape pattern of bid-ask spread and effective spread, which is consistent with Tian and Guo (2007), but conflicts with the U-shape pattern in other markets, such as NYSE (Chung et al., 1999) and Tokyo (Madhavan, 1992) markets. For the first two 15-min intervals after opening time and the last two 15-min intervals before closing time, the relative difference between bid-ask spread and effective spread is negative and statistically significant at a 1 % level, indicating that spread around the opening and closing time of the morning session are significantly greater than that of the afternoon session. It is interesting to note that the ask-bid spread is greater than the effective spread, suggesting that most transactions occur around the midpoint price (“best price”).⁶ For relative market depth, we find that the relative difference of all 15-min intervals during the morning session is significantly positive, indicating that the relative market depth in the afternoon session is greater than that in the morning session. Panel B–D exhibit a similar intraday liquidity pattern for the overnight announcement sample, lunch break announcement sample, and non-announcement sample. Overall, we find the liquidity around the opening time of the afternoon session is greater than that of the morning session.

In Panel A of Table 2, we report the return volatility across 15-min intervals during the morning session and afternoon session. We find that the relative difference of the return volatility in the first two 15-min intervals after the opening time and the last interval before the opening time is significantly negative, indicating that the return volatility around the opening time and closing time of the morning session is significantly greater than that of the afternoon session.⁷ Panel B–D in Table 2 reports the order size, trading volume, and trading frequency across 15-min intervals during the morning session and afternoon session. For order size, we find that there is a continuously increasing trend in the whole trading day. The average order size in all 15-min intervals during the morning session is significantly greater than that during the afternoon session. For trading volume and trading frequency, there is an intraday U-shape pattern. We find that trading volume and trading frequency only in the first two 15-min intervals after the opening time during the morning session are significantly greater than those during the afternoon session.

4.2. Call auction use and open price efficiency

In the preceding section, we show the periodicity of trading activities, such that there is lower liquidity around the opening time of the morning session than the afternoon session. Since our main interest in this study is finding out whether the pre-open call auction mechanism is important in determining the opening price, it is pertinent to commence our analysis by examining the price discovery and informational efficiency of opening price during morning session and afternoon session. First, we employ weighted price contribution to examine how the opening price of the morning session and the afternoon session incorporate new information from the non-trading period in the opening price. Then, we adopt unbiasedness regression to examine the informational efficiency of the opening price of the morning session and the afternoon session.

4.2.1. Weighted price contribution

Price discovery is the process through which new information is incorporated into stock prices. In this section, we extend our analysis of price discovery beyond the noisiness of price to examine the amount of new information from the non-trading period (overnight period and lunch break period) that is reflected in opening prices. Following Cao et al. (2000) and Barclay and Hendershott (2003, 2008), we use a measure called the weighted price contribution (WPC) to quantify the amount of price discovery in each period. The WPC measures the information arrival in each time period by determining the fraction of the trading session ($PreClos - Close$) stock returns that are discovered in each period. To examine the difference in opening price efficiency between the morning and afternoon sessions, we calculate the WPC for the overnight and lunch break periods respectively. The WPC is calculated at the stock level and then averaged across all stocks for each of the periods.

$$WPC_i = \sum_{t=1}^T \left(\frac{|\Delta P_t|}{\sum_{t=1}^T |\Delta P_t|} \right) \left(\frac{\Delta P_{t,t}}{\Delta P_t} \right) \quad (4)$$

⁵ We use the z-test to determine whether there is significant difference between the mean of two groups.

⁶ The similar phenomenon, the ask-bid spread is greater than the effective spread, also found in Deutsche Boerse Exchange (Riordan and Storckenmaier, 2012) and Toronto Stock Exchange (Foerster and Karolyi, 1998).

⁷ we find that the return volatility in the announcement sample is not greater than that in the non-announcement sample. The potential explanation is that there are different types of public or private information events that would be released during the non-trading period, and corporate announcement is just one kind of public information event. The other information events arriving would also shock stock returns, thus the return volatility in the announcement samples is not significantly greater than the entire sample.

where the first term of WPC is the weighting factor of each stock, and the second term is the relative contribution of the return during the period i to the total return that trading session. For the morning session, WPC_i is the weighted price contribution of period i to the total price change of the morning session. ΔP_t is the total price change of the morning session on day t from the previous afternoon closes to the morning closes. $\Delta P_{i,t}$ is the price change for period i during the morning session on day t . For the afternoon session, ΔP_t is the total price change of the afternoon session on day t from the previous morning closes to the afternoon session closes, $\Delta P_{i,t}$ is the price change for period i during the afternoon session on day t .

The WPC normalizes the proportion of preclose-to-close price evolution discovered for different periods across the trading session, such that the WPCs sum to one. The WPC is normally calculated based on a stock-by-stock basis and then averaged out across stocks. However, given that stock correlations, due to the common constituents, complicate statistical inferences about the mean, we calculate the mean WPC for each day and use the time-series standard error of the daily WPCs for statistical inference based on the spirit of [Fama and MacBeth \(1973\)](#).

In this section, to test whether the call auction mechanism accelerates new information incorporated in the opening price, we mainly focus on how much information is incorporated into the opening price by examining the weighted price contribution of the

Table 3
The weighted price contribution.

Period	PreClose to Open	Open to Open+30 m	Open+30 m to Open+60 m	Open+60 m to Open+90 m	Open+90 m to Open+120 m (Close)
Panel A: Entire Sample					
Morning Session	0.2243	0.3991	0.1567	0.1082	0.1118
Afternoon Session	0.008190	0.2577	0.2085	0.2269	0.2980
Diff	0.2161***	0.1413***	−0.05182***	−0.1187***	−0.1862***
t-value	(47.48)	(20.96)	(−9.70)	(−21.34)	(−31.33)
Panel B: Overnight Announcement Sample					
Morning Session	0.3504	0.3617	0.1195	0.08309	0.08538
Afternoon Session	0.007345	0.2607	0.2119	0.2258	0.2935
Diff	0.3430***	0.1010***	−0.09246***	−0.1427***	−0.2081***
t-value	(80.06)	(16.07)	(−18.67)	(−27.54)	(−37.53)
Panel C: Lunch Break Announcement Sample					
Morning Session	0.1728	0.4214	0.1699	0.1108	0.1217
Afternoon Session	0.01393	0.4402	0.1637	0.1518	0.2243
Diff	0.1589***	−0.0188	0.006188	−0.04099***	−0.1027***
t-value	(13.77)	(−1.28)	(0.55)	(−4.31)	(−10.32)
Panel D: Non-Announcement Sample					
Morning Session	0.1937	0.4087	0.1656	0.1141	0.1178
Afternoon Session	0.008281	0.2543	0.2086	0.2280	0.3001
Diff	0.1854***	0.1545***	−0.0430***	−0.1139***	−0.1823***
t-value	(38.72)	(22.16)	(−7.79)	(−19.88)	(−29.57)

This table reports the weighted price contribution to price change during the morning and afternoon sessions. The weighted price contribution is calculated for five different periods from close time to 120 min subsequent to the open time. For the morning session, the *PreClose* denotes the market closes on the afternoon session of the previous trading day (15:00 pm); the *Open* denotes the market opens on the morning session (09:30 am); the *Close* denotes the market closes on the morning session (11:30 am). For the afternoon session, the *PreClose* denotes the market closes in the morning session (11:30 am); the *Open* denotes the market opens in the afternoon session (13:00 pm); The *Close* denotes the market closes in the afternoon session (15:00 pm). For each period k , the weighted price contribution is obtained for each day and then averaged across days:

$$WPC_i = \sum_{t=1}^T \left(\frac{|\Delta P_t|}{\sum_{t=1}^T |\Delta P_t|} \right) \left(\frac{\Delta P_{i,t}}{\Delta P_t} \right)$$

For the morning session, ΔP_t is the price change of the morning session for day t from the previous afternoon session closes to the morning session closes. $\Delta P_{i,t}$ is the total price change for period i during the morning session on day t . For the afternoon session, ΔP_t is the price change from the previous morning closes to the afternoon session closes, $\Delta P_{i,t}$ is the price change for period i during the afternoon session on day t . Panel A reports the results for the entire sample. Panel B reports the results for the overnight announcement sample, which consists of firm-days with at least one announcement released on the overnight period and without an announcement released on any other period. Panel C reports the results for the lunch break announcement sample, which consists of firm-days with at least one announcement released on the lunch break period and without an announcement released on any other period. Panel D reports the results for the non-announcement sample, which consists of firm-day without announcement releasing on that day. *, **, and *** represent 10 %, 5 %, and 1 % significance levels, respectively.

non-trading period (overnight period or lunch break period) to the total price change of the trading session (morning session or afternoon session). For comparison purposes, we also calculate WPC for additional four half-hour periods during the trading sessions.

Table 3 reports the WPCs of the morning and afternoon sessions for the entire sample in Panel A, the overnight sample in Panel B, the lunch break sample in Panel C, and the non-announcement sample in Panel D. Two primary results emerge from this analysis. First, as Panel A shows, for the morning session, 22.43 % preclose-to-close price discovery occurs in the overnight period. For the afternoon session, only 0.819 % preclose-to-close price discovery occurs during the lunch break period. Meanwhile, the difference in WPC on the non-trading period between morning and afternoon sessions is statistically significant at a 1 % level. These findings suggest that opening prices of morning sessions incorporate more new information (from overnight), compared to opening prices of afternoon sessions (from the lunch breaks). Panel B–D provide similar evidence for the overnight, lunch break, and non-announcement samples. Opening prices of morning sessions are more informative than opening prices of afternoon sessions.

Further, Panel B reports the results for the overnight announcement sample. We find that 35.04 % of the price discovery process occurs during the overnight period, with less in the entire sample and non-announcement sample. Panel C reports the results for the lunch break announcement sample. We find that 1.393 % of the price discovery process occurs during the lunch break period, and 44.02 % occurs during the first half-hour after the afternoon session opens. However, less price discovery process occurs in the entire and non-announcement samples, consistent with corporate announcements facilitating price discovery.

Overall, the pattern of price discovery in Table 3 shows that the amount of price discovery at the opening time of the morning session is much higher than that of the afternoon session. This finding provides empirical evidence to support that the call auction mechanism could help more new information incorporated in the opening prices, and accelerate the price discovery process at the opening time.

4.2.2. Unbiasedness regressions

In this section, our main interest is to examine whether the pre-open call auction mechanism has resulted in a more informational efficient opening price. Following Biais et al. (1999) and Baruch et al. (2017), we use unbiasedness regressions, a regression method to test the extent to which stock prices reflect noise or information, to measure any changes during our sample period in the informational efficiency of the opening price. As discussed above, the pre-open call auction period accelerates more information incorporated into the opening price. As a result, it is expected that there will be less noise in the opening price of the morning session with the pre-open call auction mechanism.

We estimate the noisiness and the informational efficiency of the stock price using unbiasedness regressions. For each stock and each 1-min time period, we regress the preclose-to-close return ($Ret[PreClose, Close]$) on the return from the preclose to t minutes after opening time. ($Ret[PreClose, Open + t]$):

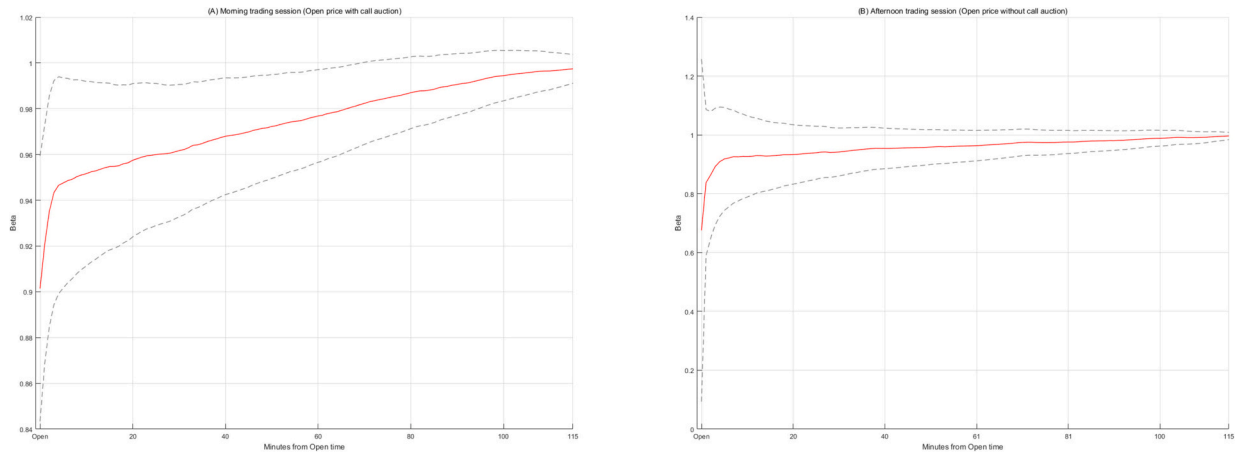


Fig. 1. The evolution of signal-to-noise ratio.

This figure presents the evolution of the signal-to-noise ratio based on the slope coefficients from unbiasedness regressions, following Biais et al. (1999). For each 1-min time period (t) in the interval Minutes [PreClose, Close], we performed regression analysis using the following regression model:

$$Ret[PreClose, Close] = \alpha + \beta Ret[PreClose, Open + t] + \varepsilon_t$$

where $Ret[PreClose, Close]$ denotes the return in the period [PreClose, Close], which is calculated as the natural logarithm of the market closing price of the trading session over the closing price of the previous trading session. $Ret[PreClose, Open + t]$ denotes the return in the period [PreClose, Open + t], which is calculated as the natural logarithm of the price at t minutes after opening time over the closing price of the previous trading session. All 1-min event intervals are defined relative to the event interval 0. For the afternoon session, event interval 0 starts from the closing time of the morning session and ends at the opening time of the afternoon.

Table 4
Unbiasedness regression.

	Open	Open+1 m	Open+2 m	Open+3 m	Open+4 m	Open+5 m	Open+30 m	Open+60 m	Open+90 m
Panel A: Entire Sample									
Morning session	0.9014	0.9206	0.9352	0.9434	0.9465	0.9476	0.9617	0.9768	0.9906
Afternoon session	0.6754	0.8733	0.8637	0.8925	0.9091	0.9182	0.9418	0.9634	0.9805
Diff	0.2260***	0.08292***	0.07146***	0.05090***	0.03748***	0.02936***	0.01992***	0.01345***	0.01008***
t-vale	(18.18)	(8.68)	(8.38)	(6.53)	(5.12)	(4.15)	(4.04)	(3.94)	(4.16)
Panel B: Overnight Announcement Sample									
Morning session	0.9184	0.9264	0.9335	0.9378	0.9381	0.9392	0.9599	0.9761	0.9908
Afternoon session	0.6168	0.8908	0.8994	0.9228	0.9357	0.9470	0.9457	0.9627	0.9799
Diff	0.3015***	0.03567*	0.03408**	0.01506	0.002460	−0.007794	0.01419*	0.01333**	0.01093***
t-vale	(10.29)	(1.73)	(2.19)	(0.056)	(0.43)	(−0.44)	(1.91)	(2.44)	(3.43)
Panel C: Lunch Break Announcement Sample									
Morning session	0.9416	0.9427	0.9503	0.9584	0.9573	0.9523	0.9697	0.9774	0.9880
Afternoon session	0.7844	0.7576	0.8809	0.9479	0.8949	0.8986	0.8758	0.9397	0.9531
Diff	0.1572***	0.1851***	0.06940***	0.01052***	0.06247***	0.05369***	0.09385***	0.03774***	0.03483***
t-vale	(2.64)	(14.95)	(14.83)	(13.99)	(12.17)	(10.39)	(3.85)	(3.52)	(7.26)
Panel D: Non-Announcement Sample									
Morning session	0.8691	0.9001	0.9231	0.9354	0.9409	0.9425	0.9573	0.9746	0.9896
Afternoon session	0.6631	0.8464	0.8775	0.9093	0.9249	0.9324	0.9425	0.9641	0.9810
Diff	0.2060***	0.05363***	0.04565***	0.02613***	0.01599*	0.01010**	0.01480***	0.01052***	0.00867***
t-vale	(15.99)	(4.86)	(4.59)	(2.87)	(1.89)	(2.21)	(2.83)	(2.91)	(3.37)

This table reports the estimation results of the slope coefficient β^* from unbiasedness regressions, following [Biais et al. \(1999\)](#). For each 1-min time period (t) in the interval Minutes [PreClose, Close], we performed regression analysis using the following regression model:

$$Ret[PreClose, Close] = \alpha + \beta Ret[PreClose, Open + t] + \varepsilon_t$$

where $Ret[PreClose, Close]$ denotes the return in the period [PreClose, Close], which is calculated as the natural logarithm of the market closing price of the trading session over the closing price of the previous trading session. $Ret[PreClose, Open + t]$ denotes the return in the period [PreClose, Open + t], which is calculated as the natural logarithm of the price at t minutes after opening time over the closing price of the previous trading session. All 1-min event intervals are defined relative to the event interval 0. For the afternoon session, event interval 0 starts from the closing time of the morning session and ends at the opening time of the afternoon session. For the morning session, the event interval 0 starts from the closing time of the afternoon session on the previous trading day and ends at the opening time of the morning session. Panel A reports the results for the entire sample. Panel B reports the results for the overnight announcement sample, which consists of firm-day with at least one announcement released on the overnight period and without announcement released on any other period. Panel C reports the results for the lunch break announcement sample, which consists of firm-day with at least one announcement released on the lunch break period and without announcements released on any other period. Panel D reports the results for the non-announcement sample, which consists of firm-day without announcement released on that day. *, **, and *** represent 10 %, 5 %, and 1 % significance levels, respectively.

$$Ret[PreClose, Close] = \alpha + \beta Ret[PreClose, Open + t] + \varepsilon_t \quad (5)$$

Although [Biais et al. \(1999\)](#) refer to this method as “unbiasedness regressions”, the slope coefficient (β) in Eq. (5) has a natural interpretation as signal-to-noise ratio. Consider the standard errors-in-variables problem for regression models ([Maddala, 1988](#)). If we assume that stock returns are serially uncorrelated and measured without error, the slope coefficient in Eq. (5) would be equal to one. Further, suppose that the “true” value of an asset is unobservable, the observable return is the combination of the “true” return and some noise element. The noise element in observable return can be related to microstructure effects such as bid-ask spread or temporary pricing error. Thus, we assume that $Ret[PreClose, Close] = RET[PreClose, Close] + \nu$ and $Ret[PreClose, Open + t] = RET[PreClose, Open + t] + \mu$, where $RET[PreClose, Close]$ and $RET[PreClose, Open + t]$ are the “true” returns. The noise elements ν and μ have zero mean and variance equal to σ_ν^2 and σ_μ^2 , respectively. An ordinary least squares estimation of Eq. (5) produces an estimated slope coefficient β^* , (where)

$$\beta^* = \beta \left(\frac{\sigma_{Ret[PreClose, Open+t]}^2}{\sigma_{Ret[PreClose, Open+t]}^2 + \sigma_\mu^2} \right) \quad (6)$$

where $\sigma_{Ret[PreClose, Open+t]}^2$ capture the total observable information from the previous close to t minute after opening time and σ_μ^2 is the noise effect in prices at t minute after opening time. The term in parentheses can be viewed as the signal-to-noise ratio. Thus, the extent to which the slope is less than one. Although this method cannot measure the signal and noise components separately, the slope coefficient β^* allows us to infer the signal-to-noise ratio.

In the spirit of [Fama and MacBeth \(1973\)](#), the unbiasedness regressions are estimated cross-sectionally for each day and each 1-min period. Then, we calculate the average regression coefficient for each 1-min period and use the time-series standard error of the mean for statistical inference. The mean slope coefficients of unbiasedness regressions are presented in [Fig. 1](#) for each minute between the opening time (9:30 a.m. for the morning session and 13:00 p.m. for the afternoon session) and 120 min after opening time, as well as the upper and lower bounds of 5 % confidence intervals. Panel A in [Fig. 1](#) presents the evaluation of the slope coefficient β^* for the morning session, the signal-to-noise ratio is high, starting at about 0.9014 at opening time (9:30 a.m.) and increasing slightly as the close (11:30 a.m.) approaches. By 11:20 a.m., the ratio. it approximately approaches one and remains at one at the reminder of the morning session. Panel B in [Fig. 1](#) presents the evolution of the slope coefficient β^* in the afternoon session, we observed a similar pattern with panel A, but the signal-to-noise starts at 0.6754 at opening time (13:00 p.m.).

Panel A in [Table 4](#) reports the estimation results of the slope coefficient β^* for the entire sample. Consistent with [Fig. 1](#), the signal-to-noise at opening time is 0.9014 for the morning session and 0.6754 for the afternoon session, and the difference is statistically significant at a 1 % level. This result indicates that the opening price of the morning session is more informational efficient (less noisy) than the afternoon session, which supports our expectation that the pre-open call auction mechanism could improve the informational efficiency of the opening price. We also find that the signal-to-noise for the first 5 min after the opening time in the morning session is significantly larger than that in the afternoon session, suggesting that the trading price at the first 5 min in the morning session is more informational efficient (less noisy) than those in the afternoon session. Panel B–D provide similar evidence in the overnight announcement sample, lunch break announcement sample, and non-announcement sample that the opening price of the morning session is more informationally efficient than that of the afternoon session.

Overall, these results show that the opening price of the morning session is much more efficient than that of the afternoon session. Our findings provide empirical evidence to support that the pre-open call auction mechanism could accelerate price discovery and improve the informational efficiency of the opening price, which is consistent with [Comerton-Forde and Rydge \(2006\)](#) and [Ibikunle \(2015\)](#).

4.3. Overnight announcements vs. lunch break announcements

So far, we have demonstrated that pre-open call auctions play an important role in improving opening price efficiency and incorporating more non-trading period information into the opening price. However, how the pre-open call auction mechanism influences the market reaction to corporate announcements is still unclear. In this section, we compare how the morning session reacts to the overnight announcement and how the afternoon session reacts to the lunch break announcement. Specifically, we first examine the price discovery process following overnight and lunch break announcements, then, we examine the effect of overnight and lunch break announcements on market liquidity.

4.3.1. Price discovery following overnight and lunch break announcements

Given that the number of lunch break announcements is much less than that of overnight announcements, there is a selection issue regarding lunchtime announcements. Considering this, instead of a [Heckman \(1976\)](#) adjustment, we use a matched sample of lunch break and overnight announcements. Therefore, we need to account for the fundamental difference between overnight announcements and lunch break announcements in terms of the information content, the type of announcement, the market environment, and the overlap issue. Prior literature documents that there is a delayed market reaction to corporate announcements in the Chinese stock market ([Chu et al., 2023](#); [Kim et al., 2019](#)), thus, two-day period returns are used to measure the information content of corporate

Table 5

Lunch break announcements and matched overnight announcements.

Period	Firms	Firm-day	Percentage	Mean Ret	Std Ret	Median Ret
Panel A: Total Announcement Events						
Lunch Break	1673	10,901	1.00	0.004761	0.04005	0.005476
Matched Overnight	1673	10,901	1.00	0.004735	0.03978	0.005513
Panel B: Positive Announcement Events						
Lunch Break	1475	6320	0.5608	−0.02914	0.02849	−0.01968
Matched Overnight	1474	6320	0.5608	−0.02902	0.02828	−0.01963
Panel C: Negative Announcement Events						
Lunch Break	1355	4581	0.4392	0.02934	0.02723	0.02104
Matched Overnight	1355	4581	0.4392	0.02920	0.02694	0.02098

This table reports the number and proportion of lunch break announcements and matched overnight announcements and the descriptive statistics for the two-day period return (Ret). We follow the one-to-one matching principle to obtain the matched overnight announcement, and Appendix B reports the detail matching principle. The positive (negative) announcement is the announcement with the positive (negative) return over the two-day period.

announcements. It is calculated as the natural logarithm of the closing price on the day after the trading day associated with the announcement, divided by the closing price on the day before the trading day associated with the announcement. The detailed matching principle is presented in Appendix B. For each lunch break announcement, we follow a one-to-one matching principle to find its matched overnight announcement. When ensuring the quality of matches, only 10,901 lunch break announcements find their matched overnight announcements.

Panel A in Table 5 reports the distribution of the lunch break announcements and their matched overnight announcements. A total of 10,901 lunch break announcements are released on 10,901 firm-days from 1673 firms, with the same for the matched overnight announcement sample. The mean two-day period return of the lunch break announcement sample is 0.4761 %, and that of the matched overnight announcement sample is 0.4733 %. The difference in the two-day period return between the lunch break announcement sample and the matched overnight announcement sample is 0.59 %, which is less than the matching principle (10 %), suggesting that the lunch break announcement sample and the matched overnight announcement sample are well-matched. Considering the asymmetry in responses to positive versus negative information (Soroka, 2006), we further divide our sample into two subsamples, positive versus negative. The lunch break announcements and their matched overnight announcements are interpreted as positive or negative based on the return over a two-day period. An announcement with a positive (or negative) return over a two-day period is considered a positive (or negative) announcement. Panel B and Panel C in Table 5 report the summary description of the positive announcement sample and negative announcement sample, respectively. As Panel B–C in Table 5 show, 56.08 % of 6320 announcements of lunch break announcements and their matched overnight counterpart are positive, with the remaining 43.92 % being negative (4949 announcements).

Although we follow the one-to-one matching principle to address the issue of sample selection bias and ensure the quality of sample selection, the multiple announcement features and corporate characteristics make it difficult to find a perfectly matched overnight announcement for each lunch break announcement. Fortunately, as Table 5 shows, the overnight announcement sample and lunch break announcement sample are well-matched, and we believe that the influence of the sample selection bias is limited in our empirical analysis.

To assess the difference in the speed of price discovery between lunch break announcements/afternoon sessions and matched overnight announcements/morning sessions, we track the intraday returns on a portfolio with long positions in stocks corresponding to firms with positive announcements and short positions in stocks corresponding to firms with negative announcements on the event date. We use a 5-min event interval. Given that morning (afternoon) sessions on Chinese stock exchanges operate between approximately 09:30 am and 11:30 am (13:00 pm and 15:00 pm), we divide each trading session into 24 intervals. For the lunch break announcements, event interval 0 starts at the closing time of the morning session (11:30 am) and ends at the opening time of the afternoon session (13:00 pm). All other intervals are defined relative to the event interval 0, with the event interval + 1 starting at the opening time of the afternoon session (13:00 pm) and ending at 13:05 pm. All subsequent intervals are five minutes in duration. Similarly, for the matched overnight announcements, event interval 0 starts at the closing time of the afternoon session (15:00 pm.) in the previous trading day and ends at the opening time of the morning session (09:30 am.). The event interval + 1 starts at the opening time (09:30 am.) of the morning session and ends at 09:35 am. Other intervals are defined relative to the event interval, with all are five minutes in duration.

For each interval, we calculate the excess returns for each stock as the difference between its return and the CSI 300 index return, where the stock and index returns are based on the closing price in each event interval. We then calculate the return of two equal-weighted portfolios: one for lunch break announcements and another for the matched overnight announcements.

Table 6 reports the two portfolios' returns in 30 five-minute intervals from −6 to +24. For the lunch break announcements, the portfolio returns are positive and statistically significant for the intervals from 0 to +24, consistent with the market taking at least

Table 6

Portfolio returns for the lunch break and matched overnight announcements.

Event Interval	Matched Overnight Announcements			Lunch Break Announcements		
	Mean Return (%)	t-stats	Cumulative return (%)	Mean Return (%)	t-stats	Cumulative return (%)
-6	-0.008338	-1.78*	-0.008338	-0.007929	-1.81*	-0.007929
-5	-0.003090	-0.65	-0.01143	-0.002335	-0.60	-0.01026
-4	0.006122	1.27	-0.005306	-0.004570	-1.16	-0.01483
-3	0.001229	0.26	-0.004077	0.001122	0.27	-0.01371
-2	-0.008697	-1.83*	-0.01277	-0.005076	-1.28	-0.01879
-1	-0.009696	-2.91***	-0.02247	0.002963	0.70	-0.01583
0	0.7382	20.14***	0.7157	0.01726	11.53***	0.001432
1	0.3003	25.07***	1.0160	0.4383	32.94***	0.4397
2	0.1699	18.73***	1.1859	0.04631	9.41***	0.4861
3	0.1139	14.92***	1.2997	0.04195	9.21***	0.5280
4	0.1007	14.10***	1.4004	0.03813	7.95***	0.5661
5	0.07789	12.35***	1.4783	0.04574	10.11***	0.6119
6	0.07373	12.54***	1.5520	0.03398	7.79***	0.6458
7	0.05437	9.53***	1.6064	0.04176	9.88***	0.6876
8	0.04474	8.64***	1.6512	0.05214	12.26***	0.7398
9	0.05116	10.33***	1.7023	0.03999	9.66***	0.7797
10	0.03122	6.15***	1.7335	0.04372	10.62***	0.8235
11	0.04511	9.41***	1.7787	0.04932	12.07***	0.8728
12	0.03553	7.76***	1.8142	0.04471	10.53***	0.9175
13	0.04043	8.92***	1.8546	0.02956	6.89***	0.9471
14	0.02983	6.83***	1.8844	0.03071	7.76***	0.9778
15	0.02512	5.90***	1.9096	0.04329	10.44***	1.0211
16	0.03541	8.64***	1.9450	0.05012	12.42***	1.0712
17	0.03417	7.65***	1.9792	0.05808	13.53***	1.1293
18	0.03607	8.57***	2.0152	0.04981	11.59***	1.1791
19	0.02096	4.88***	2.0362	0.06029	13.33***	1.2394
20	0.03528	8.58***	2.0715	0.06567	14.03***	1.3050
21	0.03080	7.25***	2.1023	0.05996	13.01***	1.3650
22	0.03601	8.49***	2.1383	0.05993	13.59***	1.4249
23	0.02911	6.91***	2.1674	0.07869	17.05***	1.5036
24	0.05137	12.30***	2.2188	0.05047	13.71***	1.5541

This table reports the portfolio returns around the releasing time of lunch break and matched overnight announcements on the trading strategy of buying stocks with positive announcement news and selling stocks with negative announcement news. For lunch break announcements, event interval 0 starts at the closing time of the morning session and ends at the opening time of the afternoon Session. For matched overnight announcements, event interval 0 starts from the closing time of the afternoon session on the previous trading day and ends at the opening time of the morning session. The other event intervals defined relative to the event interval 0: six 5-min intervals prior and twenty-four 5-min intervals subsequent to the event interval 0. The t-statistics present the test of the null hypothesis of the zero mean portfolio return. *, **, and *** represent 10 %, 5 %, and 1 % significance levels, respectively.

twenty-four five-minute intervals to absorb the information content of lunch break announcements.⁸

Furthermore, the return for interval 0 is lowest at 0.01726 % and the return for interval + 1 is highest at 0.4383 %, consistent with the majority of the price adjustments occurring during the first five-min interval (interval + 1), rather than during the lunch break (interval 0). Similarly, portfolio returns in response to matched overnight announcements are positive and statistically significant for all twenty-four intervals after interval 0. However, the return for interval 0 is highest at 0.7382 %, falling gradually in the subsequent interval. This is consistent with most price adjustments occurring during the overnight period. In terms of magnitude, the portfolio returns in interval 0 are greater, on average, for matched overnight announcements than for lunch break announcements. Therefore, we infer that pre-open call auctions prior to morning sessions accelerate price adjustments to announcements.

Fig. 2 presents respectively the detailed evolution tendency of the cumulative returns for lunch break announcements and their matched overnight announcements. Together with the results in Table 6, we find that although price adjustments to both lunch break announcements and their matched overnight announcements occur continuously during afternoon sessions and morning sessions, more information from the overnight announcements is incorporated into prices more quickly during morning sessions. More importantly, the change of the cumulative returns at the even interval 0 reflects the price reactions to public information at opening time. As Fig. 2 shows, increases at the event interval 0 appear more substantial in response to matched overnight announcements than

⁸ The price discovery process of corporate announcements continues for an extended period of time. Contrary to efficient market hypothesis (EFM), a corporate announcement does not lead to a full, instantaneous adjustment of stock prices, but to a slow drift. The potential reason for the delay price adjustment to corporate announcements is that investors usually under-react to the value-relevant information from announcements or they process the information with a significant delay, which is well-established in prior literature about post-earnings announcement drift (Huang et al., 2015; Hirshleifer et al., 2009).

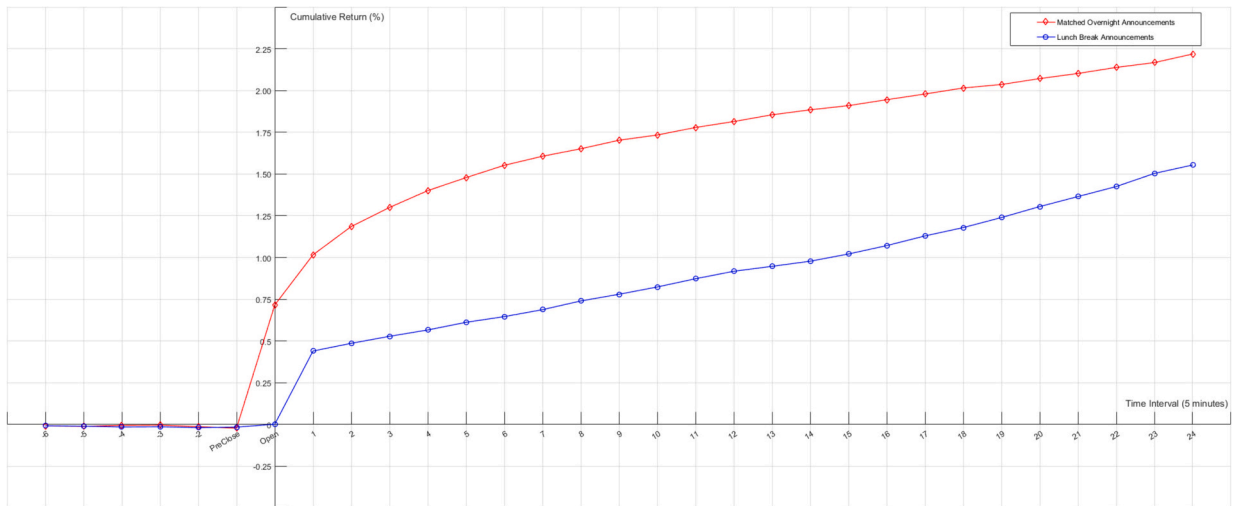


Fig. 2. The Cumulative portfolio returns for the lunch break and matched overnight announcements.

This figure presents the cumulative portfolio returns of lunch break announcements and matched overnight announcements. This figure presents the mean cumulative portfolio returns of six 5-min intervals prior to and 24 5-min intervals subsequent to the event interval 0 for a trading strategy of buying stocks with positive announcement news and selling stocks with negative announcement news. All lunch break announcements are released during the lunch break period from 11:30 am. to 13:00 pm., and for each lunch break announcement, we follow the one-to-one matching principle to find its matched overnight announcement, which is released after the market closes of the previous trading day and before the market opens. For lunch break announcements, event interval 0 starts from the closing time of the morning session and ends at the opening time of the afternoon session. For matched overnight announcements, the event interval 0 starts from the closing time of the afternoon session on the previous trading day and ends at the opening time of the morning session.

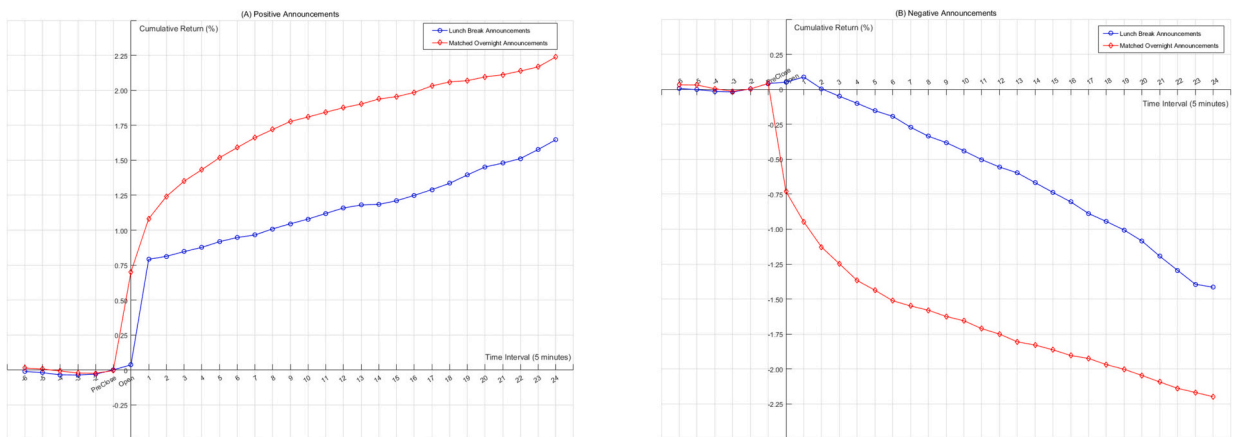


Fig. 3. The cumulative returns for the lunch break and matched overnight announcements: Positive vs. Negative announcements.

This figure presents the cumulative returns of lunch break announcements and matched overnight announcements. This figure presents the mean cumulative returns of six 5-min intervals prior to and 24 5-min intervals subsequent to the event interval 0. All lunch break announcements are released during the lunch break period from 11:30 am. to 13:00 pm., and for each lunch break announcement, we follow the one-to-one matching principle to find its matched overnight announcement, which is released after the market closes of the previous trading day and before the market opens. For lunch break announcements, event interval 0 starts from the closing time of the morning session and ends at the opening time of the afternoon session. For matched overnight announcements, the event interval 0 starts from the closing time of the afternoon session on the previous trading day and ends at the opening time of the morning session. Panel A presents the mean cumulative returns for positive announcements. Panel B presents the mean cumulative returns for negative announcements.

to lunch break announcements. This suggests that the pre-open call auctions facilitate more information content from the overnight announcements being absorbed into opening prices of the morning session. By comparison, we document that the pre-open call auctions play an important role in accelerating the price discovery process and improving opening price efficiency. This finding is consistent with Moshirian et al. (2012), despite differing market mechanism designs.

Fig. 3 presents the evolution of price discovery for positive and negative announcements, respectively. Similar to Fig. 2, no matter whether positive announcements or negative announcements, the price adjustment of the morning session to overnight announcements is faster than that of the afternoon session to lunch break announcements, especially at the opening time. Furthermore, Fig. 3 also suggests that the price adjustment to negative versus positive lunch break announcements is asymmetry, the price discovery for negative lunch break announcements is faster than that for positive lunch break announcements, especially at the opening time. However, there is no significant asymmetry in responses to positive versus negative overnight announcements.

Overall, by comparing the price adjustment with respect to overnight announcements versus the lunch break announcements, we show that overnight announcements are incorporated more quickly into stock prices upon the commencement of trading relative to lunch break announcements. Specifically, the price adjustment at the opening time of the morning session is significantly greater than that of the afternoon session. This finding suggests that the pre-open call auction plays an important role in accelerating price discovery when a corporate announcement is released.

4.3.2. The effect of the corporate announcement on market liquidity

Having established that pre-open call auctions play an important role in accelerating price adjustment to announcements, we further examine the effect of corporate announcements on market liquidity and how pre-open call auctions affect this liquidity dynamic. We perform pooled-sample analyses using interaction variables based on the following regression model:

$$Liq_{i,t,k} = \alpha + \beta_1 D_{-S_{i,t,k}} + \beta_2 D_{-A_{i,t,k}} + \beta_3 D_{-S_{i,t,k}} \times D_{-A_{i,t,k}} + \beta_4 Volt_{i,t,k} + \beta_5 \ln(Volume_{i,t,k}) + \beta_6 \ln(OS_{i,t,k}) + \beta_7 \ln(TF_{i,t,k}) + \beta_8 Ret_{i,t,k} + \varepsilon \quad (7)$$

where $Liq_{i,t,k}$ is the liquidity of stock i during the first 30-min period after the opening time of session k ⁹ on day t . We use three proxies for market liquidity, quote spread (QS), efficient spread (ES), and relative market depth (Depth). $D_{-S_{i,t,k}}$ is a session dummy variable, which is equal to one for afternoon sessions and to zero for morning sessions. $D_{-A_{i,t,k}}$ is an announcement dummy variable, equal to one if at least one announcement is released during the non-trading period prior to session k and zero if no announcement is released. We also control for return volatility (Volt), trading volume (Volume), order size (OS), trading frequency (TF), and return (Ret) during the first 30 min after opening times of each session.

The session dummy variable (D_{-S}) allows us to determine whether there is any difference in liquidity between the morning and afternoon sessions. The announcement dummy variable (D_{-A}) allows us to determine whether there is any difference in liquidity between announcements being released and no announcements being released. The interaction between D_{-S} and D_{-A} allows us to determine whether there is any difference in liquidity impacts between what ensues from overnight announcements versus lunch break announcements.

Panel A in Table 7 reports the regression results of estimating Eq. (7) using three specifications. In Columns 1–3, liquidity is measured by quote spread (QS), efficient spread (ES), and relative market depth (Depth), respectively. We are interested in the coefficients of D_{-S} , D_{-A} , and the interaction between D_{-S} and D_{-A} . Among the specifications (1–3), the coefficients of D_{-S} are -0.0000225 for QS, -0.0001519 for ES, and 0.02435 for Depth, they are all statistically significant at a 1 % level.

These results suggest that the liquidity during the first 30-min period after the opening time of the morning session is significantly greater than that of the afternoon session. This is consistent with our findings reported in Table 1. We find that the coefficients of D_{-A} are significantly negative for QS and ES, and significantly positive for Depth, consistent with corporate-announcement arrivals enhancing liquidity during the first 30 min period after the opening time of trading sessions. It is important to note that the interaction terms between D_{-S} and D_{-A} are significantly positive for both QS and ES, and significantly negative for Depth. These results show that the enhancing effect on stock liquidity is significantly greater for overnight announcements than for lunch break announcements.

We conduct additional robustness checking on the above findings, examining liquidity during the entire trading session. Panel B of Table 7 reports similar results to Panel A. Overall, we show that corporate announcements improve liquidity during the whole trading session. The enhancing effect on the liquidity of overnight announcements is greater than for lunch break announcements.

Collectively, we observe that the market reactions are significantly different in overnight announcements and lunch break announcements due to the pre-open call auctions. The findings provide empirical evidence to support that the pre-open call auctions play a key role in market reactions to announcements by accelerating price discovery and improving liquidity.

5. Conclusions

Like the Tokyo stock market, trading days in the Chinese stock markets consist of two trading sessions, the morning session and the afternoon session. These are split by a lunch break. However, in the Chinese stock market, unlike in Tokyo, pre-open call auctions operate only prior to morning sessions. These pre-open call auctions determine opening prices. However, there is no such price-setting auction prior to afternoon sessions.

⁹ There are two trading sessions on each trading day, Session 1 refers to the morning session and Session 2 refers to the afternoon session.

Table 7

The effect of the corporate announcement on market liquidity.

	<i>QS</i>	<i>ES</i>	<i>Depth</i>
Panel A: First 15-min period			
<i>D_S</i>	−0.0000225*** (−5.17)	−0.0001519*** (−25.61)	0.02435*** (17.34)
<i>D_A</i>	−0.0000174*** (−16.99)	−0.00000660*** (−5.69)	0.009023*** (25.00)
<i>D_S</i> × <i>D_A</i>	0.0000208*** (17.86)	0.00000744*** (5.70)	−0.01288*** (−21.881)
<i>Volt</i>	1.2307*** (516.51)	0.6871*** (402.01)	15.6933*** (7.68)
<i>ln(Vol)</i>	0.0001292*** (26.34)	0.0000539*** (18.63)	−0.003206** (−1.99)
<i>ln(OS)</i>	−0.0000369*** (−16.31)	0.000037*** (18.41)	0.002941*** (3.81)
<i>Ln(TF)</i>	−0.0002473*** (−50.27)	−0.0001254*** (−47.18)	−0.04489*** (−21.97)
<i>Ret</i>	−0.0009862*** (−33.35)	−0.0004604*** (−20.36)	0.1446*** (23.56)
<i>Constant</i>	0.0007064*** (15.96)	−0.0001686*** (−8.11)	0.2932*** (15.34)
<i>N</i>	3,452,575	3,452,575	3,452,575
<i>F</i>	5713.1	3063.8	764.86
<i>R2</i>	0.7862	0.5725	0.1687
Panel B: Entire trading session			
	<i>QS</i>	<i>ES</i>	<i>Depth</i>
<i>D_S</i>	−0.0000329*** (17.69)	−0.0000183*** (−12.23)	0.0009865*** (13.40)
<i>D_A</i>	−0.0000211*** (−30.16)	−0.00000895*** (−19.78)	0.0003150*** (11.57)
<i>D_S</i> × <i>D_A</i>	0.0000213*** (34.11)	0.00000994*** (22.66)	−0.0001461 (−5.39)
<i>Volt</i>	1.2331*** (265.68)	0.6499*** (275.94)	2.14*** (5.91)
<i>ln(Vol)</i>	0.0002185*** (36.39)	0.0001084*** (35.43)	0.004129*** (15.80)
<i>Ln(OS)</i>	−0.0000154*** (−7.65)	0.0000175*** (14.57)	0.0007227*** (7.80)
<i>Ln(TF)</i>	−0.0003288*** (−54.95)	−0.0001606*** (−53.97)	−0.09783*** (−30.42)
<i>Ret</i>	−0.0005663*** (−52.51)	−0.0002425*** (−42.71)	0.002341*** (6.28)
<i>Constant</i>	−0.0000907 (−1.63)	−0.0004012*** (−15.84)	0.009084*** (3.17)
<i>N</i>	3,461,231	3,461,231	3,461,231
<i>F</i>	2272.4	1751.9	658.98
<i>R2</i>	0.8316	0.7768	0.2362

This table reports the estimation results of the regression analysis investigating the effect of the lunch break announcement and overnight announcement on the market liquidity during the subsequent trading session. Our sample consists of all firm-days without any announcement and firm-days with at least one lunch break announcement or overnight announcement. The dependent variable is three proxies for market liquidity, quote spread, effective spread, and quote depth. The trading session dummy variable (*D_S*) is equal to 1 if the trading session is the afternoon session and is equal to 0 if the trading session is the morning session. The announcement dummy variable (*D_A*) is equal to 1 if there is at least one announcement released before the trading period, and is equal to 0 if there is no announcement released before the trading session. Panel A reports the effect of the corporate announcement on the market liquidity during the first 15-min period after the opening time of the trading session. Panel B effect of the corporate announcement on the market liquidity during the whole morning session or afternoon session. The t-values are reported in parentheses below the coefficient estimates. Statistical significance at the 10 %, 5 %, and 1 % levels are denoted by *, **, and ***, respectively.

This unique design of the Chinese stock market allows us to examine whether pre-open call auctions are important to price efficiency and to the price discovery process by comparing aspects of market functioning between morning sessions and afternoon sessions. We focus on the impact of pre-open call auctions on price efficiency and price discovery.

Examining the impact of pre-open call auctions on opening price efficiency, we find that the design of pre-open call auctions is important to enhance opening price efficiency. When comparing the amount of price discovery at opening times of trading sessions, we find that opening prices of morning sessions incorporate more information from the non-trading period than that of the afternoon session. Unbiasedness regression analysis implies that the opening price of the morning session contains less noise and more informational efficiency than that of the afternoon session.

Then, we attempt to explore the influence of the pre-open call auction mechanism on the market reaction to corporate announcements by examining the price and liquidity reaction of the morning session to overnight announcements and the afternoon session to lunch break announcements. To better compare the difference in the price adjustment between overnight announcements and lunch break announcements, we identify matched overnight announcements for each lunch break announcement according to a one-to-one matching principle. By comparing the return following lunch break announcements and their matched overnight announcements, we find that the price adjustments in response to overnight announcements are faster than those to lunch break announcements. More importantly, the price adjustment in response to overnight announcements primarily occurs during the pre-opening period of the morning session.

Further, we examine the impact of corporate announcements on stock liquidity. Our results show that both overnight announcements and lunch break announcements enhance stock liquidity during subsequent trading sessions. However, the enhancing effect of overnight announcements is stronger than it is for lunch break announcements, consistent with pre-open call auctions being important to stock liquidity.

Taken together, our research helps us better understand how market design features interact to influence price efficiency. We demonstrate that pre-open call auctions are a very important channel for improving opening price efficiency, facilitating the price discovery process, and enhancing stock liquidity. Our results are rich and significant for theory and hence contribute to enhancing the existing knowledge about the importance of pre-open call auctions for market functioning. Our research should have strong policy implications. Acknowledging the influential role of pre-open call auctions in advancing market efficiency, policymakers should consider strategies to further amplify their pivotal involvement in fostering price discovery. First, our study is relevant for policymakers interested in the pre-open call auction introduction, which may support policymakers in their decision-making and help bring benefits to the general public. Second, policymakers should pay more attention to the timing of the release of corporate public information. In addition, its implications should be considered valuable and informative for investors in providing practical insights to help develop better investment strategies.

Despite the incremental contributions of our study to academic research and practice, we acknowledge several limitations and propose avenues for future exploration. First, data confidentiality policies restrict our access to order data in the pre-opening call auction period. This limits our ability to capture the underlying mechanisms of the pre-opening call auction period improving price efficiency. Second, while sample-matching, on the whole, has controlled announcement characteristics, the evolution of the market environment inevitably leads to interference in the comparison between the overnight announcements and lunch break announcements.

In future research, this work can be extended in several possible directions. One potential idea could be to examine the determinants of the opening price and analyze the price discovery process during the pre-open call auction period in more detail. Also, it would be interesting to take a deeper look at the trading behavior in the pre-open call auction period. The presented analysis might provide additional evidence to help practitioners better understand the mechanism of price formation. The presented properties and determinants of the price in the pre-opening call auction period may be considered in the investment decision.

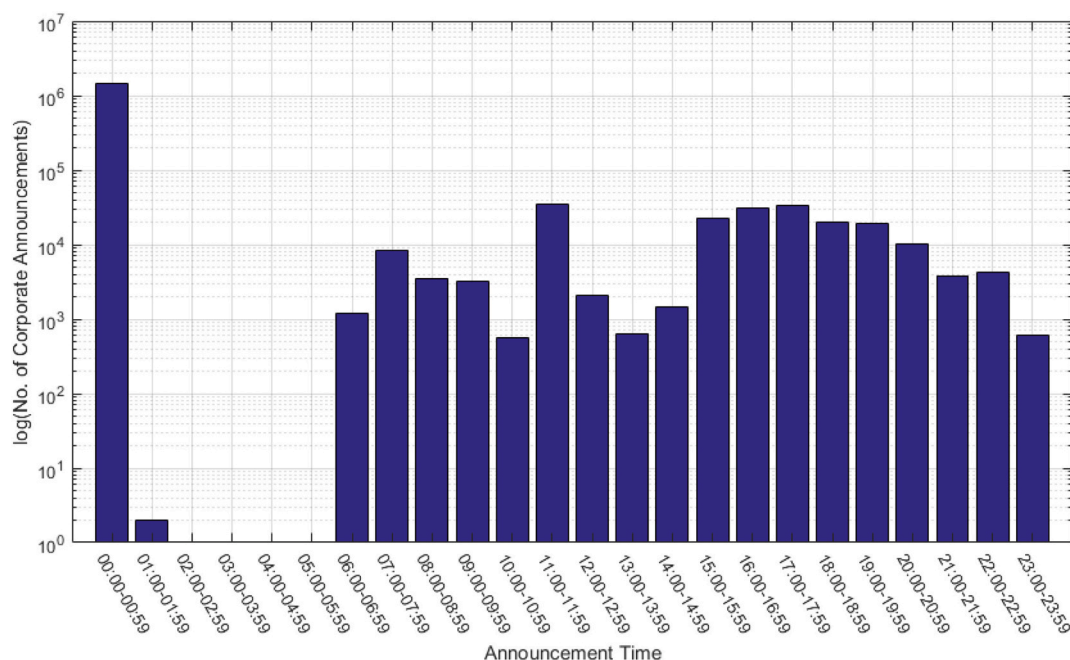
CRediT authorship contribution statement

Gang Chu: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **John W. Goodell:** Writing – review & editing, Writing – original draft, Supervision, Project administration. **Xiao Li:** Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

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Appendix A. Appendix

**Fig. A1.** Temporal distribution of corporate announcements

This figure presents the natural logarithm of the number of corporate announcements in our sample at one-hour intervals. For the Chinese stock market, there are 1,637,764 corporate announcements published in the period from January 1, 2013 to October 31, 2017, including 1,598,458 (97.60 %) announcements released during the overnight period (15:00–23:59 and 00:00–09:30), 3255 (0.199 %) announcements released during the normal trading period (09:30–11:30 and 13:00–15:00), and 36,051 (2.201 %) announcements released during the lunch break period (11:30–13:00).

Table A1

Characteristics of corporate announcements

Panel A: Number of Announcements in different periods					
Periods	N	%	Firms	Firm-days	Effective Firm-days
Overnight period	1,598,458	97.600 %	3562	553,347	355,294
Lunch break period	36,051	2.201 %	2033	22,068	17,769
Trading period	3255	0.199 %	976	2865	2865

Panel B: Announcement Types						
Announcement Types	Lunch break period		Overnight period		Trading Period	
	N	%	N	%	N	%
Initial Public Offerings (IPO)	11	0.0305 %	225	0.0141 %	0	0.000 %
Periodical Reports	263	0.729 %	98,838	6.183 %	0	0.000 %
Seasoned Equity Offerings (SEO)	2040	5.6589 %	50,542	3.162 %	2	0.0614 %
Convertible Bond	53	0.147 %	2504	0.157 %	0	0.000 %
Other Financing	262	0.727 %	6440	0.403 %	0	0.000 %
Equity distribution and Restricted share listing	291	0.807 %	38,207	2.390 %	0	0.000 %
Shareholding Changes	1523	4.225 %	35,501	2.221 %	0	0.0000 %
Major Transactions	5009	13.894 %	169,113	10.579 %	8	0.246 %
Shareholder Meeting	2469	6.849 %	155,294	9.715 %	3	0.0922 %
Investor Relations' Information	177	0.491 %	7968	0.499 %	2496	76.682 %
Clarification, Risk Warning and Performance Notice	1253	3.476 %	50,189	3.139 %	2	0.0614 %
Other Major Issues	15,337	42.543 %	562,025	35.160 %	28	0.860 %
Special Treatment and Delisting	7	0.0194 %	708	0.0443 %	0	0.0000 %
Complementary and Correcting Bulletins	826	2.291 %	12,917	0.808 %	1	0.0307 %
Intermediate Agencies' Reports	3017	8.369 %	174,686	10.928 %	0	0.000 %
Listed Company System	2528	7.012 %	172,774	10.809 %	0	0.000 %
Funds' Periodic Reports	0	0.000 %	3	0.0002 %	0	0.000 %

(continued on next page)

Table A1 (continued)

Panel B: Announcement Types						
Announcement Types	Lunch break period		Overnight period		Trading Period	
	N	%	N	%	N	%
Funds' Interim Reports	6	0.0166 %	12	0.001 %	0	0.000 %
Bonds' Announcements	97	0.269 %	14,980	0.937 %	0	0.000 %
Exchange and Clearing Companies' Announcements	294	0.816 %	2961	0.185 %	107	3.287 %
Others	588	1.631 %	42,571	2.663 %	608	18.679 %
Total	36,051	100.000 %	1,598,458	100.000 %	3255	100.000 %

This table reports the announcements characteristic in the Chinese stock market for January 1, 2013 to October 31, 2017. We collect all corporate announcements from the CSMAR dataset. Panel A reports the number and proportion of announcements released in the overnight period, lunch break period, and regular trading period, respectively. Note that the effective firm-days denote the normal trading firm-days. Panel B summarizes the announcement types in the Chinese stock market. Following the classification of the CSMAR database, we list 21 announcement types for all corporate announcements.

Appendix B. Sample matching

To assess the impact of pre-open call auctions on the speed of price adjustment to corporate announcements, we should keep relatively homogenous between the lunch break announcement sample and overnight announcement sample both in type and information content. Given the number of lunch break announcements are relatively less, we follow the one-to-one matching principle to obtain the matched overnight announcement based on the feature of lunch break announcements. The matching principle is as follows:

The matching principle is as follows:

1. To avoid any overlap with other announcements, we exclude lunch break announcements released on days with other announcements. We also exclude lunch break announcements if any announcements are released on the day before or after the day associated with the lunch break announcement.
2. Lunch break announcements and their matched overnight announcements are associated with the same listed firm.
3. Matched overnight announcements should be released on a day without any other announcements. There is no announcement released on the day before and after that day associated with the matched overnight announcement.
4. The type of the matched overnight announcement should be the same as the lunch break announcement.
5. The lunch break announcement has the same information content as the matched overnight announcement. We use the return over a two-day period, which is calculated as the natural logarithm of the close price on the day after the trading day associated with the announcement divided by the close price on the day before that trading day, to measure the information content of the announcement. To achieve as many matches as possible and ensure the quality of matches, the difference in information content between the lunch break announcement and its overnight announcement should be within the error range of 10 %.
6. Both the lunch break announcement and its matched overnight announcement should be both positive or both negative news.
7. To control the market environment, the time interval of the lunch break announcement and its matched overnight announcement should be less than 18 months.

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